Obelisk Proof Assistant — Reference Demonstration

Variant: Higher-Order Logic & Morse–Kelley Class Theory • Author: Georgy Dunaev • contact@obelisk.mozmail.com

1) Propositional logic

1.01) Axm. A01 (Affirmation of Consequent):

(α → (β → α))

Syntax side condition: α, β : Bool

1.02) Axm. A02 (Distribution of Implication):

((α → (β → γ)) → ((α → β) → (α → γ)))

Syntax side condition: α, β, γ : Bool

1.03) Axm. A03 (Conjunction Elimination Left):

((α ∧ β) → α)

Syntax side condition: α, β : Bool

1.04) Axm. A04 (Conjunction Elimination Right):

((α ∧ β) → β)

Syntax side condition: α, β : Bool

1.05) Axm. A05 (Conjunction Introduction):

(α → (β → (α ∧ β)))

Syntax side condition: α, β : Bool

1.06) Axm. A06 (Disjunction Introduction Left):

(α → (α ∨ β))

Syntax side condition: α, β : Bool

1.07) Axm. A07 (Disjunction Introduction Right):

(β → (α ∨ β))

Syntax side condition: α, β : Bool

1.08) Axm. A08 (Disjunction Elimination / Cases):

((α → γ) → ((β → γ) → ((α ∨ β) → γ)))

Syntax side condition: α, β, γ : Bool

1.09) Axm. A09 (Negation Introduction / Contradiction):

((¬α) → (α → ⊥))

Syntax side condition: α : Bool

1.10) Axm. A10 (Reductio Ad Absurdum):

((α → β) → ((α → ¬β) → ¬α))

Syntax side condition: α, β : Bool

1.11) Axm. A11 (Ex Falso Quodlibet):

(⊥ → α)

Syntax side condition: α : Bool

1.12) Axm. A12 (Law of Excluded Middle):

(α ∨ ¬α)

Syntax side condition: α : Bool

1.13) Rule Modus Ponens (MP):

α, (α → β) ⊢ β

Syntax side condition: α, β : Bool

2) Predicate logic

2.01) Axm. Discourse domain is non-empty:

∃(K : Class) ⊤

Syntax side condition: K : Class

2.02) Rule Forall Instantiation (FI):

((∀(x : Class) φ) → φ[t/x])

Syntax side condition: FFI(t, v_x, φ)

2.03) Rule Exists Instantiation (EI):

(φ[t/x] → ∃(x : Class) φ)

Syntax side condition: FFI(t, v_x, φ)

2.04) Rule Forall Abstraction (FA / Bernays):

(ψ → φ) ⊢ (ψ → ∀(x : Class) φ)

Syntax side condition: (NFV(v_x, ψ) ∧ NFV(v_x, Γ))

2.05) Rule Exists Abstraction (EA / Bernays):

(φ → ψ) ⊢ (∃(x : Class) (φ → ψ))

Syntax side condition: (NFV(v_x, ψ) ∧ NFV(v_x, Γ))

2.06) Rule Generalisation (Gen):

φ ⊢ ∀(x : Class) φ

Syntax side condition: NFV(v_x, Γ)

3) Set/class-theoretic foundations of mathematics

3.01) Axm. Class Comprehension Axiom:

∀(X : Class) ((X ∈ { x | φ(x) }) ⟷ (isSet(X) ∧ φ(X)))

Syntax side condition: NFV(v_C, φ)

3.02) Axm. Extensionality Axiom:

∀(A, B : Class) ((∀(x : Class) ((x ∈ A) ⟷ (x ∈ B))) → (A = B))

Syntax side condition: A, B, x : Class

3.08) Axm. Heredity Axiom (Subclasses of Sets are Sets):

∀(A, B : Class) ((isSet(B) ∧ (A ⊆ B)) → isSet(A))

Syntax side condition: A, B : Class

3.12) Axm. Regularity (Foundation) Axiom:

∀(A : Class) ((A ≠ ∅) → ∃(x : Class) ((x ∈ A) ∧ ((x ∩ A) = ∅)))

Syntax side condition: A, x : Class

3.13) Axm. Pairing Axiom:

∀(a, b : Class) ((isSet(a) ∧ isSet(b)) → isSet({a, b}))

Syntax side condition: a, b : Class

3.14) Axm. Union Axiom:

∀(A : Class) (isSet(A) → isSet(⋃ A))

Syntax side condition: A : Class

3.22) Def. IF-THEN-ELSE-FI (Class conditional with NFV side condition and context turnstile):

W, Γ ⊢ (⟦IF ψ THEN A ELSE B FI⟧ = { v | ((ψ ∧ (v ∈ A)) ∨ (¬ψ ∧ (v ∈ B))) })

Syntax side condition: (NFV(v_v, ψ) ∧ NFV(v_v, Γ))

3.23) Def. simple_if (polymorphic non-dependent conditional over arbitrary Type):

simple_if : ∏(α : Type). (Bool → (α → (α → α)))

(⟦simple_if(α, true, t₁, t₂)⟧ = ⟦t₁⟧), (⟦simple_if(α, false, t₁, t₂)⟧ = ⟦t₂⟧)

Syntax side condition: α : Type, t₁ : α, t₂ : α

3.24) Def. dependent_if (polymorphic dependent conditional with type computation):

dependent_if : ∏(T : (Bool → Type)) (b : Bool). (T(true) → (T(false) → T(b)))

(typeof(⟦dependent_if(T, b, t₁, t₂)⟧) := ⟦simple_if(Type, b, typeof(t₁), typeof(t₂))⟧)

(⟦dependent_if(T, true, t₁, t₂)⟧ = ⟦t₁⟧), (⟦dependent_if(T, false, t₁, t₂)⟧ = ⟦t₂⟧)

Syntax side condition: T : (Bool → Type), b : Bool, t₁ : T(true), t₂ : T(false)

3.32) Thm. Ordinal Induction (∈-Ind):

((∀(α : Class) ((α ∈ On) → ((∀(β : Class) ((β ∈ α) → φ(β))) → φ(α)))) → ∀(α : Class) ((α ∈ On) → φ(α)))

Syntax side condition: (Fresh(v_β, v_α) ∧ NFV(v_β, φ(α)))

3.33) Def. Power Class Definition (𝒫(a)):

∀(a, x : Class) ((x ∈ 𝒫(a)) ⟷ (isSet(x) ∧ (x ⊆ a)))

Syntax side condition: Fresh(v_x, v_a)

3.34) Axm. Power Set Axiom (Power Class of a Set is a Set):

∀(a : Class) (isSet(a) → isSet(𝒫(a)))

Syntax side condition: a : Class

3.37) Axm. Replacement / Limitation of Size Axiom:
∀(F, A : Class) ((isClassFunction(F) ∧ isSet(A)) → isSet(F[A]))
Syntax side condition: F, A : Class

3.40) Def. Dunaev Tagged Ordered Pair:
∀(A, B : Class) ([A, B]_D = { z | ∃(a ∈ A) (z = {{{{a}}}, {∅}}) ∨ ∃(b ∈ B) (z = {{{{b}}}, {∅, {∅}}}) })
Syntax side condition: A, B : Class

3.41) Thm. First Projection of Dunaev Tagged Ordered Pair:
∀(A, B : Class) ({ a | ({{{{a}}}, {∅}} ∈ [A, B]_D) } = A)
Syntax side condition: A, B : Class

3.42) Thm. Second Projection of Dunaev Tagged Ordered Pair:
∀(A, B : Class) ({ b | ({{{{b}}}, {∅, {∅}}} ∈ [A, B]_D) } = B)
Syntax side condition: A, B : Class

3.45) Axm. Infinity Axiom:
∃(x : Class) (isSet(x) ∧ ((∅ ∈ x) ∧ ∀(y : Class) ((y ∈ x) → ((y ∪ {y}) ∈ x))))
Syntax side condition: x, y : Class

3.46) Axm. Global Choice Axiom:
∃(c : Class) (isClassFunction(c) ∧ ∀(x : Class) ((isSet(x) ∧ (x ≠ ∅)) → (c(x) ∈ x)))
Syntax side condition: c, x : Class

3.47) Def. Universal Class (V):
(V = { x | (x = x) })
Syntax side condition: V : Class

3.53) Def. Transitive Class Predicate:
(Tr(A) ⟷ ∀(x : Class) ((x ∈ A) → (x ⊆ A)))
Syntax side condition: A, x : Class

3.66) Axm. Tarski's Axiom of Universes:
∀(x : Class) (isSet(x) → ∃(u : Class) ((isUniverse(u) ∧ (x ∈ u)) ∧ isSet(u)))
Syntax side condition: x, u : Class

3.67) Def. Semi-Singleton Class Predicate (semi_singleton):
(semi_singleton(x) ⟷ ∀(u : Class) ∀(v : Class) (((u ∈ x) ∧ (v ∈ x)) → (u = v)))
Syntax side condition: x, u, v : Class

3.68) Def. Value Extraction via Semi-Singleton Subclasses (sss / Value):
(sss(x) = ⋃({ z | ((z ⊆ x) ∧ semi_singleton(z)) }))
Syntax side condition: x, z : Class

3.69) Thm. Theorem of Evaluation of a Family ((Λx, t)(z) = t[z/x]):
∀(z : Class) ((isSet(z) ∧ isSet(t[z/x])) → (((Λ(x : Class), t))(z) = t[z/x]))
Syntax side condition: (t : Class.Expression), (v_x : Variable), (z : Class)

3.70) Def. Set-Theoretic Model of Dependent Sum (ΣB / Sigma Model):
(Σ(B) = { p | ∃(x : Class) ∃(y : Class) (((x ∈ Dom(B)) ∧ (y ∈ B(x))) ∧ (p = [x, y]_D)) })
Syntax side condition: B, p, x, y : Class

3.71) Def. Set-Theoretic Model of Dependent Product (ΠB / Pi Model):
(Π(B) = { f | ((isClassFunction(f) ∧ (Dom(f) = Dom(B))) ∧ ∀(x : Class) ((x ∈ Dom(B)) → (f(x) ∈ B(x)))) })
Syntax side condition: B, f, x : Class

4) Metalogical laws

4.01) Axm. Instantiation Axiom:
∀(A : Type) ∀(G : Ctx) ∀(φ : Fm) ∀(t : eval(A)) ∀(v : Variable) (W, G ⊢ ((∀(v : A) φ) → (let v = t in φ)))
Syntax side condition: t : eval(A), v : Variable

4.02) Axm. Generalisation Axiom:
∀(A : Type) ∀(Γ : Ctx) ∀(φ : Fm) ∀(v : Variable) (¬(Enc(v) ∈ Ctx.FV(Γ)) → ((W, Γ ⊢ φ) → (W, Γ ⊢ ∀(v : A) φ)))
Syntax side condition: NFV(v, Γ)

4.03) Rule CoNecessitation:
(F, [] ⊢ ‹›, Γ ⊢ φ›) ⊢ (F, Γ ⊢ φ)
Syntax side condition: F : Theory, Γ : Ctx, φ : Fm

4.04) Def. Quantifier Unique (!):

∀(φ : Fm) ((!(z : Class) φ) ⟷ ∀(x, y : Class) ((φ[x/z] ∧ φ[y/z]) → (x = y)))

Syntax side condition: z, x, y : Variable

4.05) Def. Quantifier Exists and Unique (∃!):

∀(φ : Fm) (T, [] ⊢ ((∃!(x : Class) φ) ⟷ ((∃(x : Class) φ) ∧ !(x : Class) φ)))

Syntax side condition: x : Variable, φ : Fm

4.06) Axm. Class Comprehension Axiom (Metatheory):

∀(φ : Fm) ∀(x : Name) ∀(k : Class.Expr) (FFI(k, x, φ) → (T, [] ⊢ ((k ∈ { x | φ }) ⟷ (isSet(k) ∧ φ[k/x]))))

Syntax side condition: FFI(k, x, φ)

4.07) Axm. Soundness / Reflection Axiom:

∀(φ : Fm) (⟦φ⟧ → (F, [] ⊢ φ))

Syntax side condition: φ : Fm, F : Theory

4.08) Axm. Theory Inclusion (T ⊆ F):

∀(Γ : Ctx) ∀(φ : Fm) ((T, Γ ⊢ φ) → (F, Γ ⊢ φ))

Syntax side condition: T, F : Theory, Γ : Ctx, φ : Fm

4.09) Def. Syntactic Encoding Operator (Enc):

∀(T : Type) ∀(s : T) (isSet(Enc(s)) ∧ (Enc(s) ∈ Emb(T)))

Syntax side condition: T : Type, s : T

4.10) Def. Syntactic Decoding & Roundtrip Law (Dec):

∀(T : Type) ∀(s : T) ((Dec(Enc(s), Emb(T)) : T) = (s : T))

Syntax side condition: T : Type, s : T

4.11) Def. Dual Type Embedding Operator (Emb):

∀(X : Class) ((Emb(Class)(X) ⟷ ⊤) ∧ (Emb(Set)(X) ⟷ isSet(X)))

Syntax side condition: X : Class

4.12) Def. Proof-Object Class Family (ProofsOf):

∀(s : Stmt) ∀(p : Prf) ((Enc(p) ∈ ProofsOf(Enc(s))) ⟷ CheckProof(p, s))

Syntax side condition: s : Stmt, p : Prf

4.13) Def. Class of Provable Statements (Theorems):

∀(s : Stmt) ((Enc(s) ∈ Theorems) ⟷ (∃(p : Class) (p ∈ ProofsOf(Enc(s)))))

Syntax side condition: s : Stmt, p : Class

5) Tactics and inference rules

5.01) Thm. Elementary Theorems:

((F, [(B ∈ x)] ⊢ ∃(x : Class) (B ∈ x)) ∧ (F, [(B ∈ x)] ⊢ isSet(B)))

Syntax side condition: B, x : Class

5.02) Rule Definition to Introduction Rule:

(F, [(B ∈ x)] ⊢ ∃(x : Class) (B ∈ x)) ⊢ (F, [(B ∈ x)] ⊢ isSet(B))

Syntax side condition: B, x : Class

5.03) Rule Deduction Rule Application:

(F, [] ⊢ ((B ∈ x) → isSet(B)))

Syntax side condition: B, x : Class

5.04) Thm. Tactics Proof Example:

F, [] ⊢ ((∃(y : Class) ∀(x : Class) (x ∈ y)) → ∀(x : Class) ∃(y : Class) (x ∈ y))

Syntax side condition: x, y : Class

6) Incompleteness-related statements

6.01) Axm. Theories (Metalevel Stratification):

((W ⊆ Q) ∧ ((Q ⊆ T) ∧ (T ⊆ F)))

Syntax side condition: W, Q, T, F : Theory

6.02) Thm. Löb's Theorem:

∀(φ : Fm) (Q, [] ⊢ ((T, [] ⊢ ((T, [] ⊢ φ) → φ)) → (T, [] ⊢ φ)))

Syntax side condition: φ : Fm

6.03) Thm. Gödel's Second Incompleteness Theorem:

Q, [] ⊢ (¬(T, [] ⊢ ⊥) → ¬(T, [] ⊢ ¬(T, [] ⊢ ⊥)))

Syntax side condition: T, Q : Theory

Appendix: NBG Formulas-as-Types Standard Library

24 closed proof terms compiled by NBGSurfaceCompiler in Poly/ML. A formula is a type; a proof program is a term inhabiting it. The Dunaev pair injectivity theorem is the class-correct replacement for Kuratowski pairing on proper classes.

Theorem	Formula (Type)	AST	Norm
empty_class_no_elements	(∀(z : Class) ((z ∈ { x | ⊥ }) → ⊥))	7	7
empty_class_is_subclass	(∀(X : Class) (∀(z : Class) ((z ∈ { x | ⊥ }) → (z ∈ X))))	8	8
universal_class_characterization	(∀(z : Class) (((z ∈ { x | ⊤ }) → isSet(z)) ∧ (isSet(z) → (z ∈ { x | ⊤ }))))	15	15
russell_class_is_not_a_set	(isSet({ x | ((x ∈ x) → ⊥) }) → ⊥)	19	32
binary_inter_comm	(∀(A : Class) (∀(B : Class) (∀(z : Class) ((z ∈ { x | ((x ∈ A) ∧ (x ∈ B)) }) → (z ∈ { x | ((x ∈ B) ∧ (x ∈ A)) })))))	26	26
binary_union_comm	(∀(A : Class) (∀(B : Class) (∀(z : Class) ((z ∈ { x | ((x ∈ A) ∨ (x ∈ B)) }) → (z ∈ { x | ((x ∈ B) ∨ (x ∈ A)) })))))	23	23
class_diff_subclass_left	(∀(A : Class) (∀(B : Class) (∀(z : Class) ((z ∈ { x | ((x ∈ A) ∧ ((x ∈ B) → ⊥)) }) → (z ∈ A)))))	10	10
class_eq_sym	(∀(X : Class) (∀(Y : Class) ((X = Y) → (Y = X))))	6	6
class_eq_trans	(∀(X : Class) (∀(Y : Class) (∀(Z : Class) (((X = Y) ∧ (Y = Z)) → (X = Z)))))	9	9
quantifier_swap	((∃(y : Set) (∀(x : Set) (x ∈ y))) → (∀(x : Set) (∃(y : Set) (x ∈ y))))	7	7
dunaev_tagged_pair_projections	(∀(A : Class) (∀(B : Class) ((π₁([A, B]_D) = A) ∧ (π₂([A, B]_D) = B))))	3	3
cantor_diagonal_theorem	(∀(R : Class) (∀(c : Set) (isSet(c) → ((((c ∈ { x | ([x, x](_D ∈ R) → ⊥) }) → ([c, c]_D ∈ R)) ∧ ([c, c](_D ∈ R) → (c ∈ { x | ([x, x](_D ∈ R) → ⊥) }))) → ⊥))))	28	50
self_in_singleton	(∀(x : Class) (isSet(x) → (x ∈ { u | (u = x) })))	8	8
self_in_succ	(∀(x : Class) (isSet(x) → (x ∈ { u | ((u ∈ x) ∨ (u = x)) })))	9	9
class_sub_succ	(∀(x : Class) (∀(z : Class) ((z ∈ x) → (z ∈ { u | ((u ∈ x) ∨ (u = x)) }))))	12	12
dunaev_pair_injective	(∀(A : Class) (∀(B : Class) (∀(C : Class) (∀(D : Class) (([A, B]_D = [C, D]_D) → ((A = C) ∧ (B = D)))))))	26	26
dunaev_cartesian_elim	(∀(A : Class) (∀(B : Class) (∀(p : Class) ((p ∈ { q | (∃(u : Set) (∃(v : Set) (((u ∈ A) ∧ (v ∈ B)) ∧ (q = [u, v]_D)))) }) → (∃(u : Set) (∃(v : Set) (((u ∈ A) ∧ (v ∈ B)) ∧ (p = [u, v]_D))))))))	9	9
inter_sub_left	(∀(A : Class) (∀(B : Class) (∀(z : Class) ((z ∈ { u | ((u ∈ A) ∧ (u ∈ B)) }) → (z ∈ A)))))	10	10
inter_sub_right	(∀(A : Class) (∀(B : Class) (∀(z : Class) ((z ∈ { u | ((u ∈ A) ∧ (u ∈ B)) }) → (z ∈ B)))))	10	10
union_sub_left	(∀(A : Class) (∀(B : Class) (∀(z : Class) ((z ∈ A) → (z ∈ { u | ((u ∈ A) ∨ (u ∈ B)) }))))	13	13
semi_singleton_sub_elim	(∀(X : Class) (∀(z : Class) ((z ∈ { w | ((w ∈ X) ∧ (∀(u : Class) (∀(v : Class) (((u ∈ X) ∧ (v ∈ X)) → (u = v))))) }) → (z ∈ X))))	9	9
family_fiber_char	(∀(B : Class) (∀(x : Class) (∀(y : Class) ((y ∈ { w | ([x, w]_D ∈ B) }) → ([x, y]_D ∈ B)))))	8	8
sigma_dependent_sum_intro	(∀(A : Class) (∀(B : Class) (∀(x : Class) (∀(y : Class) (((x ∈ A) ∧ ([x, y]_D ∈ B)) → ((x ∈ A) ∧ ([x, y]_D ∈ B)))))))	8	8
pi_dependent_product_elim	(∀(A : Class) (∀(B : Class) (∀(f : Class) (((∀(x : Class) ((x ∈ A) → ([x, f]_D ∈ B))) ∧ (∀(x : Class) (x ∈ A))) → (∀(x : Class) ([x, f]_D ∈ B))))))	10	10

Reproduce locally: python3 obelisk_cli.py compile examples/nbg_verified_programs.nbg